FIRST — THE MOST IMPORTANT CLARIFICATION

We Did Not Change the 0–3 Scores

Before anything else, this needs to be clear:

✅ **The 0–3 scoring in the database is untouched.**
Every score assigned to every (disease × risk factor) cell — whether it's a 0, 1, 2, or 3 — remains exactly as the rules engine assigned it. We read those numbers and use them. We have not changed a single cell value.

What we changed is **how we compare clinical domains against each other** — which is a completely separate layer on top of the DB scores. Think of it like this:

❌ V1 — OLD LOGIC

- Read the 0–3 scores from DB
- Sum them up per disease
- Group by clinical domain
- Take AVG per domain
- **Show top domains by raw AVG**

VS

✅ V2 — NEW LOGIC

- Read the 0–3 scores from DB ← same
- Sum them up per disease ← same
- Group by clinical domain ← same
- Take AVG per domain ← same
- **Subtract what any person of this age/gender would score, then show domains that are elevated ABOVE that baseline**

Everything up to the last step is identical. We added one calibration layer. That's it.

DRAWBACK 1 — THE MOST FUNDAMENTAL PROBLEM

Raw Scores Measure Database Size, Not Patient Risk

The raw AVG score of a domain is directly influenced by **how many diseases exist in that domain**. A domain with more diseases has more rows to collect the patient's score — so it will almost always score higher. This is a mathematical artifact, not a clinical signal.

Clinical Domain	Disease Count	Effect on Raw Score
Injury/Toxicology	1,982	Massive score collection advantage
Cardiovascular	583	Large score collection advantage
MSK/Rheumatology	425	Medium — gets dominated
ENT	151	Small — almost always buried
Sleep Medicine	79	Very small — never surfaces

👨 Patient: Ramesh, 32 years old, works night shifts, has severe sleep apnea in family

What the patient actually needs to monitor: Sleep Medicine (Sleep Apnea risk is real and documented)

❌ **V1 RESULT (RAW SCORING)**

Sleep Medicine has only 79 diseases. Even with high scores in those 79 diseases, the raw AVG is overwhelmed by Cardiovascular (583 diseases) and Injury/Toxicology (1,982 diseases) which collect the same age/gender base scores across far more rows.

⚠️ **Sleep Medicine never surfaces. False negative.**

✅ **V2 RESULT (EXCESS SCORING)**

We subtract the age/gender baseline from each domain. Sleep Medicine's excess is high because night shift work and family history specifically elevate it above baseline. Cardiovascular's excess is low because the patient has no cardiac-specific factors.

✅ **Sleep Medicine correctly flagged.**

📄 DIRECT QUOTE — DB DOCUMENTATION (ASSUMPTION A2)

"Scores 0-3 are ordinal ranks of evidence strength, NOT probabilities or odds ratios. Totals are relative within this scoring scheme and are NOT directly convertible to absolute risk (%)."

The DB's own documentation says the totals are **relative within this scheme** — meaning a Cardiovascular total of 14 vs a Sleep Medicine total of 3 does NOT mean cardiovascular is more dangerous for this patient. They are on different scales because they have different numbers of diseases. Comparing them directly is mathematically invalid.

DRAWBACK 2 — CONFIRMED BY LIVE TESTING

Family History Always Went to the Wrong Domain

The database has 11 generic family history columns: `famhist_father_rf`, `famhist_mother_rf`, `famhist_siblings_rf`, and so on. These columns are **not specific to any condition** — they simply represent "a family member is sick" with a generic heritability weight.

When any of these columns are activated, they give the same boost to every disease in the database. Now look at what happens mathematically:

```
-- All 7 family members selected for Bone/Joint Disorders -- famhist_father_rf activated, famhist_mother_rf activated, etc. Domain: Cardiovascular → 583 diseases × family boost = large total Domain: MSK/Rheumatology → 425 diseases × family boost = smaller total -- Cardiovascular wins purely because 583 > 425 -- The patient's actual concern (bones) is IGNORED Result: "Heart & Circulation: Worth Monitoring" Reality: Patient has strong bone disease family history — bones should be flagged
```

🧪 This Was Tested Live — Not Theoretical

During development, all 7 family member groups were selected for **Bone / Joint Disorders**. After clicking Recalculate:

❌ V1 RESULT

Bones & Joints showed **"Looking good"** — despite the patient having the maximum possible bone disease family history.

Heart & Circulation was flagged instead.

⚠️ **Direct false negative. Observed live.**

✅ V2 RESULT

We stripped the generic `famhist_*` columns from domain scoring entirely. Instead, we route each specific condition directly to its clinical domain.

"Father has Bone Disease" → directly boosts MSK/Rheumatology only.

✅ **Bones & Joints correctly flagged.**

📄 DIRECT QUOTE — DB DOCUMENTATION (ASSUMPTION A10)

"Ten FamHist columns (mother, father, grandparents, aunts/uncles, siblings) are scored according to how much a first-degree vs more distant relative contributes. First-degree (mother, father, siblings) typically score higher than second-degree."

Notice what this assumption does NOT say: it does not say the columns are condition-specific. The DB knows *who* the family member is (father vs grandfather) but has no idea *what condition* they have. That information only exists in our application layer — which is why we handle it there.

DRAWBACK 3 — EVERY PATIENT LOOKS THE SAME

Age & Gender Dominate Everything — Making the Profile Non-Personal

In V1, the biggest contributors to any domain's score are the age and gender columns — because these apply to every single disease. A 60-year-old's age column activates across all 16,383 diseases. Age is the loudest signal by far.

The result: **two completely different patients with the same age and gender get nearly identical risk profiles**, regardless of their actual lifestyle differences.

👥 Two Patients — Same Age, Completely Different Lives

👤 PATIENT A — HEALTHY 55-YEAR-OLD

Non-smoker. Normal BMI. No comorbidities. No medications. Exercises daily. Clean diet. No family history of anything.

👤 PATIENT B — HIGH-RISK 55-YEAR-OLD

Smoker. Obese (BMI 34). Has Diabetes + Hypertension. On 4 medications. Father and mother both had heart attacks. High salt + processed food diet.

❌ V1 RESULT — BOTH PATIENTS

Both 55-year-olds get the same top domains: Cardiovascular, Injury/Toxicology, Oncology — dominated by age column scores. The healthy patient looks almost as risky as the high-risk patient.

⚠️ **The profile is the same for both. Useless.**

✅ V2 RESULT

Patient A: Most domains show "Looking good." The excess above age baseline is minimal — correctly reflecting a healthy patient.

Patient B: Cardiovascular, Endocrine, Respiratory all flagged — the excess above baseline is massive due to smoking, obesity, diabetes, family history.

✅ **Two very different results for two very different people.**

A risk profile that gives the same result to a healthy person and a high-risk person is not a risk profile. It is an age chart.

DRAWBACK 4 — ADMITTED IN THE DB'S OWN DOCUMENTATION

Most Diseases Have Only Generic Chapter-Level Scores (Not Specific Evidence)

> 📄 DIRECT QUOTE — DB DOCUMENTATION (ASSUMPTION A11) — CRITICAL
>
> "~15,400 of 16,383 diseases have at least one non-zero cell populated via chapter-level Tier A rules. Only a few hundred of the most-prevalent/well-studied diseases have rich, condition-specific Tier B/C rules. For rare diseases outside those, the score reflects only generic chapter patterns and should be supplemented by expert review before clinical use."

This is the DB documentation admitting that **15,400 out of 16,383 diseases get their scores from broad chapter defaults** — not specific clinical evidence. The Tier A rules say things like: "all cardiovascular diseases get a high score for age 40–59" — regardless of what the specific disease is.

This means:

- The Cardiovascular domain's high raw scores are largely **chapter-level generic defaults**, not specific clinical associations
- When V1 shows "Heart & Circulation: Worth Monitoring" for a 50-year-old, it is mostly reflecting generic chapter rules applied to 583 diseases — not the patient's actual cardiac risk
- Excess scoring corrects for this by removing exactly these generic effects (age + gender baseline) and keeping only what is truly patient-specific

📌 **Simple analogy**

Imagine a school where 90% of marks are given based on which class a student is in (chapter defaults), and only 10% are based on the student's actual performance. Ranking students by total marks would mostly rank them by which class they're in — not by how well they actually performed. Excess scoring removes the "which class are you in" part and ranks by actual performance only.

DRAWBACK 5 — FIXED THRESHOLD VS. SELF-CALIBRATING

No Fair Way to Decide What "High" Means With Raw Scores

In V1, to decide which domains to flag as "Worth Monitoring," you need to answer: what raw score is high enough? This is a fixed threshold problem — and any fixed number will fail:

Scenario	V1 Fixed Threshold Problem
Patient fills in only 3 fields (age, gender, BMI)	All domain scores are low. Nothing crosses the threshold. Patient sees no flags — even though they may have real risk from just these 3 factors.
Patient fills in all 19 categories with many risk factors	All domain scores are high. Everything crosses the threshold. Patient sees 15 domains flagged — creating panic and losing meaning.
Cardiovascular raw score = 14, ENT raw score = 2	How do you know if 2 is "high for ENT" or "low for ENT"? Without knowing ENT's typical range, a fixed threshold is meaningless.

📊 How V2 Solves This — Self-Calibrating Threshold

V2 uses: **Threshold = Mean of all excess scores + 0.5 × Standard Deviation**

This threshold adapts to each individual patient's own score distribution:

HEALTHY PATIENT (FEW RISK FACTORS)

All excess scores are small and close together. Mean is low, SD is small. Threshold is low. Very few or no domains cross it. Correctly: nothing major to worry about.

HIGH-RISK PATIENT (MANY FACTORS)

Excess scores are spread wide. A few domains spike far above the mean. Threshold catches only the genuinely elevated ones. Not everything — just the real outliers.

✅ **Each patient gets a threshold calibrated to their own risk profile. No false alarms. No missed signals.**

ADDITIONAL POINT

The DB Itself Says Additivity Is a Simplification

> **DIRECT QUOTE — DB DOCUMENTATION (ASSUMPTION A1)**
>
> "Additivity: risk factors are treated as ADDITIVE, not multiplicative. This is a simplification; in reality, many risk factors interact (e.g., smoking × HTN × diabetes for CAD has a super-additive effect on MI risk). Additive scoring is defensible as a triage heuristic but should NOT be interpreted as an epidemiological probability."

The DB creators themselves acknowledge that simply summing scores is a heuristic — not a precise medical calculation. They built it as a **starting point for triage**, not as a final answer. V2's calibration layer is exactly the kind of improvement they were suggesting downstream systems should add.

> **DIRECT QUOTE — DB DOCUMENTATION (ASSUMPTION A12)**
>
> "NOT for individual medical decisions: This matrix is a triage and decision-support heuristic. It is NOT a diagnostic tool, NOT a validated clinical prediction rule, and must NOT replace clinical judgment."

The DB is designed as a heuristic foundation. Our calibration layer (excess scoring) makes the output of that heuristic meaningfully personalised — which is exactly what a patient-facing product needs.

SUMMARY

All 5 Drawbacks at a Glance

1

Database size bias

Domains with more diseases always outscore smaller domains — regardless of actual patient risk. Raw scoring measures the DB, not the patient.

2

Family history goes to wrong domain

Generic `famhist_*` columns boost all diseases equally. Cardiovascular always wins because it has the most diseases. Confirmed by live test.

3

Every patient looks the same

Age and gender dominate raw scores. A healthy 55-year-old and a high-risk 55-year-old get nearly identical profiles. No personalisation.

4

Scores are generic chapter defaults

The DB itself admits 94% of diseases use only generic chapter-level rules (Tier A). Raw scores mostly reflect chapter patterns, not patient-specific risk.

5

No fair threshold possible

Any fixed cutoff for "what is high" either flags everything or flags nothing depending on how many details the patient filled in.

THE BOTTOM LINE

Summary

💬 **Conclusion**

The V2 logic uses the exact same 0–3 scores from the database — nothing was changed there. What was added is one calibration step: subtract what any generic person of the same age and gender would score, and flag only what is elevated above that. The database's own documentation (Assumptions A1, A2, A11) confirms that raw scores are a simplified heuristic not meant for direct comparison across domains. The old logic was tested live: a patient with all 7 family members having bone disease was told their bones are fine — a direct false negative on a medical platform. V2 fixed that specific failure and all others described in this document.

Cura Health — Risk Scoring Logic Review | April 2026

Cura Health — Risk Scoring Architecture | April 2026